

Exp 1: Setting Up the Environment and Preprocessing the Data

Aim

To set up a machine learning environment and preprocess the Pet Adoption dataset by handling missing values, encoding categorical variables, performing feature scaling, and splitting the dataset into training and testing sets.

Algorithm

1. Import Required Libraries
Import NumPy, Pandas, Matplotlib, Seaborn, and required Scikit-learn modules.
2. Load the Dataset
Load the `pet_adoption.csv` dataset using Pandas.
3. Explore the Dataset
Use `head()`, `info()`, `describe()`, and `isnull().sum()` to understand the dataset.
4. Handle Missing Values
Replace missing categorical values using the mode and numerical values using the median.
5. Separate Features and Target
Separate `AdoptionSpeed` as the target variable and the remaining columns as input features.
6. Encode Categorical Variables
Convert categorical data into numerical form using `LabelEncoder`.
7. Perform Feature Scaling
Standardize numerical features using `StandardScaler`.
8. Split the Dataset
Divide the preprocessed data into training and testing sets using `train_test_split()`.
9. Display Preprocessed Data
Display the final preprocessed dataset and the training/testing data dimensions.

CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder

print("Libraries imported successfully!")
df = pd.read_csv("pet_adoption.csv")
print("\nDataset loaded successfully!")
print("Dataset Shape:", df.shape)
print("\nFirst 5 rows:")
display(df.head())
print("\nDataset Information:")
df.info()
print("\nStatistical Description:")
display(df.describe())
print("\nMissing Values:")
print(df.isnull().sum())
df["Breed"] = df["Breed"].fillna(df["Breed"].mode()[0])
df["Vaccinated"] = df["Vaccinated"].fillna(df["Vaccinated"].mode()[0])
df["Fee"] = df["Fee"].fillna(df["Fee"].median())
print("\nMissing values after preprocessing:")
print(df.isnull().sum())
X = df.drop("AdoptionSpeed", axis=1)
y = df["AdoptionSpeed"]
print("\nFeatures Shape:", X.shape)
print("Target Shape:", y.shape)
categorical_cols = X.select_dtypes(include=["str", "object"]).columns
print("\nCategorical Columns:")
print(list(categorical_cols))
X_encoded = X.copy()
label_encoders = {}
for col in categorical_cols:
    le = LabelEncoder()
    X_encoded[col] = le.fit_transform(X_encoded[col].astype(str))
    label_encoders[col] = le
print("\nCategorical encoding completed!")
print("\nEncoded Data:")
display(X_encoded.head())
numerical_cols = X_encoded.select_dtypes(include=np.number).columns
scaler = StandardScaler()
X_scaled = X_encoded.copy()
X_scaled[numerical_cols] = scaler.fit_transform(X_encoded[numerical_cols])

```

```

print("\nFeature scaling completed!")
print("\nScaled Data:")
display(X_scaled.head())
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)
print("\nDataset split completed!")
print("Training data shape:", X_train.shape)
print("Testing data shape:", X_test.shape)
print("Training target shape:", y_train.shape)
print("Testing target shape:", y_test.shape)
print("\nFinal Preprocessed Training Data:")
display(X_train.head())
print("\n=====")
print("EXPERIMENT 1 COMPLETED SUCCESSFULLY")
print("=====")
print("Total samples      :", len(df))
print("Total features     :", X_scaled.shape[1])
print("Training samples    :", len(X_train))
print("Testing samples     :", len(X_test))
print("Number of classes   :", y.nunique())

```

OUTPUT:

```
Libraries imported successfully!
```

```
Dataset loaded successfully!  
Dataset Shape: (500, 14)
```

First 5 rows:

petType	AgeMonths	Gender	Breed	Vaccinated	Dewormed	Sterilized	Health	MaturitySize	FurLength	Fee	PhotoAmount	VideoAmount	AdoptionSpeed
Dog	64	Male	Beagle	Yes	No	Yes	Healthy	Small	Medium	732.0	4	0	1
Cat	86	Male	Persian	No	Yes	No	Healthy	Large	Medium	1748.0	6	0	2
Cat	33	Female	Mixed Breed	No	Yes	Yes	Healthy	Medium	Medium	3924.0	9	2	3
Dog	88	Male	Beagle	No	Yes	Yes	Healthy	Small	Medium	423.0	10	0	3
Dog	34	Male	Mixed Breed	Yes	Yes	Yes	Healthy	Large	Short	3478.0	10	1	2

```
Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
0   PetType                500 non-null    str
1   AgeMonths              500 non-null    int64
2   Gender                 500 non-null    str
3   Breed                  495 non-null    str
4   Vaccinated             495 non-null    str
5   Dewormed                500 non-null    str
6   Sterilized              500 non-null    str
7   Health                 500 non-null    str
8   MaturitySize            500 non-null    str
9   FurLength               500 non-null    str
10  PhotoAmount             495 non-null    float64
11  PhotoAmount             500 non-null    int64
12  VideoAmount              500 non-null    int64
13  AdoptionSpeed           500 non-null    int64
dtypes: float64(1), int64(4), str(9)
memory usage: 76.8 KB
```

Statistical Description:

	AgeMonths	Fee	PhotoAmount	VideoAmount	AdoptionSpeed
count	500.000000	495.000000	500.000000	500.000000	500.000000
mean	60.968000	2538.046465	5.66400	1.440000	2.304000
std	35.249795	1497.984308	2.78543	1.121122	1.447679
min	2.000000	11.000000	1.00000	0.000000	0.000000
25%	29.750000	1186.500000	3.00000	0.000000	1.000000
50%	59.000000	2488.000000	6.00000	1.000000	2.000000
75%	93.000000	3955.500000	8.00000	2.000000	4.000000
max	120.000000	4992.000000	10.00000	3.000000	4.000000

Missing Values:

PetType	0
AgeMonths	0
Gender	0
Breed	5
Vaccinated	5
Dewormed	0
Sterilized	0
Health	0
MaturitySize	0
FurLength	0
Fee	5
PhotoAmount	0
VideoAmount	0
AdoptionSpeed	0
dtype: int64	

Missing values after preprocessing:

PetType	0
AgeMonths	0
Gender	0
Breed	0
Vaccinated	0
Dewormed	0
Sterilized	0
Health	0
MaturitySize	0
FurLength	0
Fee	0
PhotoAmount	0
VideoAmount	0
AdoptionSpeed	0
dtype: int64	

Features Shape: (500, 13)
Target Shape: (500,)

Categorical Columns:
['PetType', 'Gender', 'Breed', 'Vaccinated', 'Dewormed', 'Sterilized', 'Health', 'MaturitySize', 'FurLength']

Categorical encoding completed!

Encoded Data:

	PetType	AgeMonths	Gender	Breed	Vaccinated	Dewormed	Sterilized	Health	MaturitySize	FurLength	Fee	PhotoAmount	VideoAmount
0	1	64	1	0	1	0	1	0	2	1	732.0	4	0
1	0	86	1	6	0	1	0	0	0	1	1748.0	6	0
2	1	33	0	5	0	1	1	0	1	1	3924.0	9	2
3	1	88	1	0	0	1	1	0	2	1	423.0	10	0
4	1	34	1	5	1	1	1	0	0	2	3478.0	10	1

Feature scaling completed!

Scaled Data:

	PetType	AgeMonths	Gender	Breed	Vaccinated	Dewormed	Sterilized	Health	MaturitySize	FurLength	Fee	PhotoAmount	VideoAmount
0	1.024295	0.086101	0.949284	-1.573663	0.608164	-2.105892	0.730577	-0.425165	1.200586	0.056456	-1.212608	-0.597993	-1.285714
1	-0.976281	0.710843	0.949284	1.110242	-1.644294	0.474858	-1.368782	-0.425165	-1.239629	0.056456	-0.530260	0.120748	-1.285714
2	1.024295	-0.794218	-1.053425	0.662925	-1.644294	0.474858	0.730577	-0.425165	-0.019522	0.056456	0.931145	1.198860	0.500000
3	1.024295	0.767638	0.949284	-1.573663	-1.644294	0.474858	0.730577	-0.425165	1.200586	0.056456	-1.420133	1.558231	-1.285714
4	1.024295	-0.765820	0.949284	0.662925	0.608164	0.474858	0.730577	-0.425165	-1.239629	1.283765	0.631611	1.558231	-0.392857

Dataset split completed!
Training data shape: (400, 13)
Testing data shape: (100, 13)
Training target shape: (400,)
Testing target shape: (100,)

Final Preprocessed Training Data:

	PetType	AgeMonths	Gender	Breed	Vaccinated	Dewormed	Sterilized	Health	MaturitySize	FurLength	Fee	PhotoAmount	VideoAmount
11	1.024295	-0.055886	0.949284	-1.573663	0.608164	0.474858	0.730577	-0.425165	-0.019522	-1.170853	-1.369763	-0.238622	1.392857
21	-0.976281	-1.191781	0.949284	-0.679028	0.608164	0.474858	-1.368782	-0.425165	1.200586	1.283765	-0.488621	-0.597993	-1.285714
38	1.024295	1.221996	-1.053425	-1.573663	0.608164	0.474858	-1.368782	-0.425165	1.200586	1.283765	1.343508	0.120748	0.500000
33	-0.976281	-0.595436	0.949284	-0.231710	-1.644294	0.474858	-1.368782	-0.425165	-0.019522	0.056456	0.438189	-0.957363	-0.392857
04	-0.976281	1.534367	0.949284	-0.231710	0.608164	0.474858	-1.368782	3.147653	1.200586	-1.170853	0.260215	0.480119	1.392857

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EXPERIMENT 1 COMPLETED SUCCESSFULLY

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Total samples : 500
Total features : 13
Training samples : 400
Testing samples : 100
Number of classes : 5

RESULT:

The Pet Adoption dataset was successfully preprocessed by handling missing values, encoding categorical variables, scaling features, and splitting the dataset into training and testing sets.